DSC 258R: NATURAL LANGUAGE PROCESSING

Spring 2026

Course Syllabus

Instructional Team Information

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Course Description

Introduction to classical methods to mine and analyze real-world text data, including tokenization, stemming and lemmatization, bag-of-words classification, word embedding, language models, sentiment analysis, part-of-speech tagging, named entity recognition, sequence-to-sequence models, and possible biases and societal implications in these models.

Course Objectives

At the end of the course, students will be able to:

- Identify the different components of the NLP pipeline—preprocessing, representation learning, and model fitting,
- Identify the classical and cutting-edge approaches to fundamental NLP problems,
- Apply processing and text mining approaches at scale using off-the-shelf tools, and
- Interpret modern representation learning methods in NLP.

Course Format

DSC 258R is a 10-week, fully asynchronous course with weekly recorded lectures provided for you, along with asynchronous quizzes and assignments, offered through Canvas, Kaggle, and Gradescope. Your instructional team will also hold weekly office hours live over Zoom. You can collaborate with other students and your instructional team using the course discussion forum in Piazza.

Course Outline

Week	Topic	Outcomes	Activities
1	Tokenization, Stemming, and Lemmatization	▪ Describe why we need tokenization in both languages with and without spaces. ▪ Describe the difference and commonality between stemming and lemmatization. ▪ Describe popular NLP packages to preprocess data.	1. Watch Week 1 Lectures 2. Complete Quiz 1
2	Bag-of-Words Classification	▪ Describe the general framework for text mining and natural language processing tasks ▪ Apply bag-of-words representations to conduct text classification ▪ Describe and apply the every (implementation) detail of bag-of-words representation	1. Watch Week 2 Lectures 2. Complete Quiz 2
3	Vector Semantics and Embeddings	▪ Outline how to represent words in NLP ▪ Define lemmas, senses of words, and word relations ▪ Define the distributional hypothesis ▪ Distinguish between TF-IDF and Word2Vec embeddings ▪ Contrast neural network architectures that are used to learn word representations (skip-gram vs. Continuous Bag of Words ▪ Identify the semantic properties of embeddings	1. Watch Week 3 Lectures 2. Complete Quiz 3
4	N-Gram Language Models	▪ Define a language model and what it can be used for in practice ▪ Describe the basics of n-gram-based language modeling techniques, along their corresponding learning methods and smoothing techniques ▪ Identify and apply the different ways in which language models are evaluated (e.g., held-out likelihood/perplexity)	1. Watch Week 4 Lectures 2. Complete Quiz 4
5	Modern Neural Language Models	▪ Describe the basics of RNN architectures ▪ Describe how an RNN can be used to parameterize a language model ▪ Describe the strengths and weaknesses of neural language models vs. their classical n-gram-based counterparts	1. Watch Week 5 Lectures 2. Complete Quiz 5
6	Part-of-Speech Tagging	▪ Define and describe the semantic meaning of the POS tags	1. Watch Week 6 Lectures 2. Complete Quiz 6

Week	Topic	Outcomes	Activities
		■ Describe the concepts of sequence labeling framework and conditional random field ■ Apply popular NLP packages to get the POS tags of a given sentence	
7	Named Entity Recognition	■ Define NER and how the BIO/BIOES labeling scheme works for NER ■ Describe the unique challenges of NER in different scenarios, including low-resource and few-shot settings ■ Apply popular NLP packages to train an NER model and also how to apply an off-the-shelf NER model	1. Watch Week 7 Lectures 2. Complete Quiz 7
8	Machine Translation and Generation	■ Describe the recurrent and transformer-based architecture behind a variety of conditional language models ■ Define the learning objectives and training techniques of conditional language models ■ Define and describe the broader range of application domains these models can target ■ Identify the describe the variety of evaluation metrics used to automatically judge quality of output text	1. Watch Week 8 Lectures 2. Complete Quiz 8 3. Complete NLP Competition
9	Biased and Societal Implications in NLP Models- Part 1	■ Outline different types of harms that could result from biases ■ Define gender bias and develop awareness of how gender biases manifest in NLP models and data ■ Identify biases in NLP models and data ■ Identify racial bias and disparities in NLP models	1. Watch Week 9 Lectures 2. Complete Quiz 9
10	Biased and Societal Implications in NLP Models- Part 2	■ Apply debiasing techniques ■ Apply debiasing techniques through data augmentation ■ Apply debiasing algorithms through adjustment	1. Watch Week 10 Lectures 2. Complete Video Assignment

Learning Materials

Required: Lecture Videos

Each weekly module contains a series of lectures covering that week's topics that we recommend you watch in the order presented. Each video is of varying length, so be sure to give yourself enough time to watch all of the content during the week to prepare yourself for that week's quiz and/or assignment.

Optional: Texts

There are no required textbooks for this class. However, if you would like to pursue more advanced topics or get another perspective on the same material, you can find a list of books below, most of which are available for free online.

Speech and Language Processing 3rd ed., by Dan Jurafsky and James H. Martin. Available at: www.web.stanford.edu/~jurafsky/slp3/

Foundations of Statistical Natural Language Processing, by Chris Manning and Hinrich Schütze, MIT Press. Cambridge, MA: May 1999. Companion website available at: nl.stanford.edu/fsnlp/

Natural Language Processing, by Jacob Eisenstein, 2018: Available at: github.com/jacobeisenstein/gt- nlp-class/blob/master/notes/Eisenstein-nlp-notes.pdf

A Primer on Neural Network Models for Natural Language Processing, by Yoav Goldberg, 2015: Available at: u.cs.biu.ac.il/~yogo/nnlp.pdf

Resource Library

You can access the course's Resource Library from the home page of this course. The Resource Library contains all of the course lecture videos organized by module, papers of interest to those who want to dive deeper into some more advanced related topics, and optional books also covering more advanced topics and other perspectives on what we cover in the course.

Assessments, Evaluation, and Grading

Quizzes

Over the course of first 8 weeks of this course, you will take 8 quizzes, all between 4 and 12 questions long, with each covering one week's worth of content. Each quiz will open at 12:00am PST on Monday and be due at 11:59pm PST the following Sunday, giving you 1 week to complete the quiz. You will only have 1 attempt and 30 minutes after you begin, so be sure to pick a time where you will be undisturbed and can finish in one sitting. Late quizzes will not be accepted.

Video Assignments: Article Review

For Module 10, you will be completing a video assignment, where you will be asked to choose a paper on the topics of identifying and remediating gender bias in NLP, review that paper, then record a short video answering a series of questions about that paper, including your analysis of its results. Each assignment will be graded by your instructional team using a rubric available in Canvas. Late assignments will not be accepted.

Coding Assignment

In order to get more hands-on experience with the topics covered in this class, you will be provided with a project to complete that will provide you the opportunity to practice the skills you are learning in lecture. This project will involve participating in a competition through Kaggle, and will be available staring in Week 5 and is due at the end of Week 8.

Grading

Assignments	*% of Grade*
9 Quizzes	72% (8 quizzes, 8% each)
Article Review Video Assignment	14%
Coding Assignment	14%

Grade	*Range*
A+	97-100
A	93-96
A-	90-92
B+	87-89
B	83-86
B-	80-82
C+	77-79
C	73-76
C-	70-72
D+	67-69
D	63-66
D-	60-62
F	0-59

Course and UCSD Policies

UCSD Code of Conduct

All participants in the course are bound by the University of California Code of Conduct (https://aisc.uci.edu/students/index.php)

Netiquette

Be respectful. Be sensitive. Be aware. Effective written communication and open academic dialogue are crucial for sustaining a learning community that is respectful, considerate, relevant, creative, and thought-provoking. In an online classroom, expressions, meaning, and tone can quickly be taken out of context, making it imperative that online learners adhere to the communication guidelines below:

- Treat your classmates with respect.
- Be thoughtful and open in a discussion.
- Be aware and sensitive to different perspectives.
- Build one another up and encourage one another to succeed.

The following behavior should be avoided:

- Using insulting, condescending, or abusive words.
- Using all capital letters, which comes across as SHOUTING.
- Contacting learners or posting advertisements and solicitations.
- Posting copyrighted material.

Academic Integrity

Academic Integrity is expected of everyone at UC San Diego. This means you must be honest, fair, responsible, respectful, and trustworthy in your actions.

Lying, cheating, or other forms of dishonesty will not be tolerated because they undermine learning and the University's ability to certify students' knowledge and abilities. Thus, any attempt to get, or help another get, a grade by cheating, lying, or dishonesty will be reported to the Academic Integrity Office and result in sanctions. Sanctions can include an F in the class and suspension or dismissal from the University.

So, think carefully before you act. Before you act, ask yourself the following questions: a: is my action honest, fair, respectful, responsible, and trustworthy, and b) is my action authorized by the instructor? If you are unsure, don't ask a friend; ask your instructor, instructional assistant, or the Academic Integrity Office. You can learn more about academic integrity at academicintegrity.ucsd.edu.

Accessibility

Students requesting accommodations for this course due to a disability must provide a current Authorization for Accommodation (AFA) letter issued by the UC San Diego Office for Students with Disabilities (OSD) which is located in University Center 202 behind Center Hall. Students are required to present their AFA letters to Faculty (please make arrangements to

contact me privately) and to the OSD Liaison in the department in advance so that accommodations may be arranged. Contact the OSD for further information: https://disabilities.ucsd.edu/.

Religious Accommodation

See: EPC Policies on Religious Accommodation, Final Exams, Midterm Exams

It is the policy of the university to make reasonable efforts to accommodate students having bona fide religious conflicts with scheduled examinations by providing alternative times or methods to take such examinations. If a student anticipates that a scheduled examination will occur at a time at which his or her religious beliefs prohibit participation in the examination, the student must submit to the instructor a statement describing the nature of the religious conflict and specifying the days and times of conflict.

For final examinations, the statement must be submitted no later than the end of the second week of instruction of the quarter.

For all other examinations, the statement must be submitted to the instructor as soon as possible after a particular examination date is scheduled.

If a conflict with the student's religious beliefs does exist, the instructor will attempt to provide an alternative, equitable examination that does not create undue hardship for the instructor or for the other students in the class.

Accessibility

See: Nondiscrimination Policy Statement

CARE at the Sexual Assault Resource Center 858.534.5793 | sarc@ucsd.edu | https://care.ucsd.edu Counseling and Psychological Services (CAPS) 858.534.3755 |https://caps.ucsd.edu

Subject to Change Policy

Note that the information contained in the course syllabus, other than the grade and absence policies, may be – under certain circumstances such as a modification to enhance student learning – subject to change with reasonable advanced notice, as deemed appropriate by the instructor.